$$\begin{aligned}\sin(x+y)-\sin(x-y) &= (\sin x\cos y+\cos x\sin y)-(\sin x\cos y-\cos x\sin y)\\ &= 2\cos x\sin y\end{aligned}$$

Let $x=\dfrac{A+B}{2},\ y=\dfrac{A-B}{2} \implies x+y=A,\ x-y=B$

$$\sin A-\sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$$

$$\begin{aligned}\cos A-\cos B &= \cos\big(x+y\big)-\cos\big(x-y\big)\\ &= (\cos x\cos y-\sin x\sin y)-(\cos x\cos y+\sin x\sin y)\\ &= -2\sin x\sin y\end{aligned}$$

$$\cos A-\cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$$

Gradient of PQ:

$$\begin{aligned}m_{PQ} &= \frac{b\sin q-b\sin p}{a\cos q-a\cos p}\\ &= \frac{b}{a}\cdot\frac{2\cos\frac{p+q}{2}\sin\frac{q-p}{2}}{-2\sin\frac{p+q}{2}\sin\frac{q-p}{2}}\\ &= -\frac{b}{a}\cot\frac{p+q}{2}\end{aligned}$$

Gradient of SR:

$$\begin{aligned}m_{SR} &= \frac{b\sin s-b\sin r}{a\cos s-a\cos r}\\ &= -\frac{b}{a}\cot\frac{r+s}{2}\end{aligned}$$

$$PQ\parallel SR \iff m_{PQ}=m_{SR}$$

$$\iff \cot\frac{p+q}{2}=\cot\frac{r+s}{2}$$

$$\iff \frac{r+s}{2}=\frac{p+q}{2}+k\pi,\qquad k\in\mathbb{Z}$$

$$\iff r+s-p-q=2k\pi$$

Since

$$0 \leq p < q < r < s < 2\pi$$

$$0 < r + s - p - q < 4\pi$$

$$\therefore k = 1$$

$$\therefore PQ \parallel SR \iff r + s - p - q = 2\pi$$

This solution was submitted by **Jon**
Published and refined by **Jon**
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